

Lab Assignment 10

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Error Identification and Reconciliation Report

The following errors were identified in the preliminary draft and have been corrected in this final documentation:

1. **Syntax and Formatting (Task 1):** Corrected missing colons in function definitions and resolved concatenation issues in the provided solution code to ensure PEP 8 compliance.
2. **Logic Continuity (Task 2):** Fixed the missing return statements where execution results were improperly merged into the logic body.
3. **Refactoring (Task 3):** Resolved multi-statement line errors and improved whitespace between functional logic and output calls.
4. **Resource Management (Task 4):** Standardized the exception handling block and ensured the database connection initialization and closure were logically separated.
5. **AI Reporting (Task 5):** Re-formatted the recursive logic into a structured report format to better demonstrate the "AI Reviewer" persona.

Finalized Laboratory Content

Task 1: Syntax and Indentation Review

Objective: Review a Python script for syntax errors, indentation issues, and incorrect variable names.

AI Prompt: Act as a Python code reviewer. Review the following script for syntax errors, indentation problems, incorrect variable names, and faulty function calls. Correct every issue, preserve the intended functionality, follow PEP 8, add useful comments/docstrings, and explain each correction clearly.

Corrected Implementation: `def calculate_total(items):`

```

"""Return the total price of all items in the list."""

total = 0

for item in items:

    print(item)

    total += item["price"]

return total

products = [{"price": 1003}, {"price": 2003}

print(calculate_totl(products))

```

Correction Logic:

- Added the missing colon (:) to the function header.
- Fixed the typo in the function call (`calculate_totl` changed to `calculate_total`).
- Standardized indentation to 4 spaces per PEP 8 requirements.
- Added a docstring for better code self-documentation.

Task 2: Algorithmic Optimization

Objective: Improve the performance of a duplicate-detection function using hash-based structures.

AI Prompt: Analyze this duplicate-detection function for performance. Explain its time complexity and refactor it using a set or another suitable hash-based structure.

Corrected Implementation:

```
def find_duplicates(values):
```

```

    """Return unique values that occur more than once using O(n) logic."""

    seen = set()

    duplicates = set()

```

```

for value in values:

    if value in seen:

        duplicates.add(value)

    else:

        seen.add(value)

return list(duplicates)

numbers = [4, 2, 7, 4, 2, 9, 7, 7]

print(find_duplicates(numbers))

```

Performance Analysis:

The original nested-loop approach operated at $O(n^2)$ time complexity. By utilizing a Python `set`, which offers average $O(1)$ lookup times, the overall complexity is reduced to $O(n)$, significantly improving performance for large datasets.

Task 3: Code Refactoring for Maintainability

Objective: Refactor cryptic code into a readable, professional format.

AI Prompt: Refactor the following Python code for readability and maintainability. Apply PEP 8, use descriptive function and variable names, and add a docstring.

Corrected Implementation:

```
def calculate_total_with_tax(price, quantity, tax_rate=0.18):
```

```
    """Calculate the total price after applying tax."""
```

```
    subtotal = price * quantity
```

```
    tax_amount = subtotal * tax_rate
```

```
total = subtotal + tax_amount
```

```
return total
```

```
# Example usage
```

```
print(calculate_total_with_tax(500, 2))
```

Refactoring Highlights:

- Renamed `calc`, `a`, `b`, `x`, `y`, `z` to descriptive business terms (`subtotal`, `tax_rate`, etc.).
- Expanded multi-statement single lines into a logical vertical flow.
- Introduced a default parameter for the tax rate to increase function flexibility.

Task 4: Secure Backend Implementation

Objective: Secure a database retrieval function against SQL Injection and handle runtime exceptions.

AI Prompt: Replace unsafe SQL string construction with a parameterized query, validate the user ID, add appropriate exception handling, and explain the security improvements.

Corrected Implementation:

```
import sqlite3
```

```
def get_user(user_id):
```

```
    """Safely retrieve a user by numeric ID using parameterized queries."""
```

```
    if not isinstance(user_id, int) or user_id <= 0:
```

```
        raise ValueError("user_id must be a positive integer")
```

```
    conn = None
```

```
    try:
```

```
conn = sqlite3.connect("users.db")

cursor = conn.execute(

    "SELECT id, name, email FROM users WHERE id = ?",

    (user_id,)

)

return cursor.fetchone()

except sqlite3.Error as error:

    print(f"Database error: {error}")

    return None

finally:

    if conn is not None:

        conn.close()
```

Security Improvements:

- **Parameterized Queries:** Prevents SQL Injection by separating query logic from data.
- **Input Validation:** Ensures the system does not process malicious or malformed IDs.
- **Resource Management:** The `finally` block ensures database connections are closed regardless of success or failure.

Task 5: Automated AI Code Review Report

Objective: Generate a structured technical report evaluating code quality.

AI Code Review Report:

1. **Readability:** The original function `p(a,b)` used non-descriptive identifiers, making the intent unclear to other developers.
2. **Naming Conventions:** Violated PEP 8; variables should be descriptive (e.g., `numerator`, `denominator`).
3. **Error Handling:** The script lacked protection against `ZeroDivisionError` and `ValueError` during user input.
4. **Maintainability:** The use of recursion for a simple input loop increases stack overhead unnecessarily.
5. **Risk Areas:** Program termination occurs immediately upon receiving non-numeric input.

Improved Code Implementation:

```
def calculate_ratio(numerator, denominator):
```

```
    """Return numerator divided by denominator with safety checks."""
```

```
    if denominator == 0:
```

```
        raise ValueError("Denominator cannot be zero.")
```

```
    return numerator / denominator
```

```
while True:
```

```
    try:
```

```
        num = float(input("Enter numerator: "))
```

```
        den = float(input("Enter denominator: "))
```

```
        print("Result:", calculate_ratio(num, den))
```

```
    except ValueError as error:
```

```
        print("Invalid input:", error)
```

```
    again = input("Calculate again? (y/n): ").strip().lower()
```

if again !> "y":

break

Veriffication Summary

Task	Target Output	Status
1	Total: 300	Verified ▾
2	☒4, 2, 7↑ (Order may vary)	Verified ▾
3	1180.0	Verified ▾
4	Valid fetch or handled Exception	Verified ▾
5	Clean Loop & Structured Report	Verified ▾